Venkata Lakshmi Parimala Pasupuleti

Full-Stack Engineer · Backend-Heavy · Python · Django · Celery · PostgreSQL · Data Ingestion Pipelines · Scalable APIs · AWS

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PROFESSIONAL SUMMARY

Backend-heavy full-stack engineer with **9+ years of Python** and 7+ years shipping production systems on a 100K+ MAU multi-tenant SaaS platform. At Entrata, owned features **end-to-end**: designed backend architecture, built **data ingestion pipelines** across 1.6M+ records, wrote **Django and FastAPI services**, wired up **Celery async task queues** (idempotency, retries, dead-letter), optimized **200+ PostgreSQL queries (30% gain)**, and shipped to production independently without handoffs. At Intuit, now builds backend services with Python/FastAPI, integrates AI features, and ships via GitHub Actions CI/CD to AWS. Comfortable **owning the full stack**: Python backends, React/TypeScript frontends, REST APIs, database design, async workers, and cloud infrastructure. Small-team DNA: **120+ Agile sprints**, 300+ PR reviews, junior engineer mentoring. Stack: **Python, Django, FastAPI, Celery, PostgreSQL, RabbitMQ, React, TypeScript, Node.js, AWS, Docker**. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026).

TECHNICAL SKILLS

Backend: Python (9+ yrs), Django, FastAPI, Flask, Java (Spring Boot), Node.js, Celery, RabbitMQ, Kafka, REST APIs, GraphQL, OAuth2 & JWT, Microservices, Async Task Queues, Idempotency & Retry Logic, Pub/Sub

Data & Databases: PostgreSQL, MySQL, Snowflake, Redis, MongoDB, SQLite, ETL/ELT Data Ingestion Pipelines, Data Modeling, Query Optimization, Stored Procedures, AWS Glue, Data Quality Monitoring

Frontend & Full-Stack: React, Redux, TypeScript, JavaScript, PHP, Laravel, HTML/CSS, REST API Design, GraphQL

Cloud & DevOps: AWS (S3, EC2, RDS, Lambda, SQS, Glue, API Gateway, IAM), Docker, Kubernetes, Terraform, GitHub Actions CI/CD, Prometheus, Grafana, Linux, HIPAA/PHI RBAC, Libsodium Encryption

WORK EXPERIENCE

Intuit

Jun 2026 – Present

Software Engineer — Backend · Python/FastAPI · Data Pipelines · Financial Analytics

- Build backend Python/FastAPI services for financial data processing; own feature delivery end-to-end from technical scoping through production deployment; build **ETL/ELT pipelines on Snowflake** from data ingestion and feature engineering through prediction APIs (scikit-learn, XGBoost); containerize with Docker and ship via **GitHub Actions CI/CD to AWS** with Prometheus observability.
- Integrate AI capabilities into backend services (Anthropic Claude); implement guardrails, input validation, and fallbacks; build LLM evaluation harnesses (golden datasets + CI regression); **collaborate directly with product and business stakeholders** to translate requirements into shipped features; maintain structured audit trails and data quality monitoring.

Entrata — Multifamily & Affordable Housing SaaS

Jan 2017 – Jul 2024

Senior Software Engineer — Backend · Django · Celery · PostgreSQL · Data Pipelines

- Built production backend systems in **Python (Django, FastAPI), PHP (Laravel), Java, and React** for a 100K+ MAU multi-tenant SaaS platform; **designed and owned data ingestion pipelines** processing 1.6M+ HUD/Tax Credit financial records; built **Celery async task queues** (pub/sub, idempotency, retry logic, dead-letter handling) for rent updates across thousands of tenant records at **99.5% uptime**; **optimized 200+ PostgreSQL queries (30% performance gain)** across 1M+ records.
- Owned features **end-to-end** across 120+ Agile sprints: architecture, backend API development, React/TypeScript frontends, PostgreSQL schema design, async workers, and AWS deployment; **built 56 state-specific certification documents** and automated workflows eliminating 80% of manual workload; **Docker microservices** redesign (50% better scalability, 30% faster deployments); AWS S3 migration with Libsodium encryption and RBAC for PHI — 100% compliance; reviewed 300+ PRs, mentored junior engineers.

PROJECTS

AgentProbe — Open-Source Backend Platform (15-Endpoint FastAPI)

Backend · Python · APIs

Python · FastAPI · Celery · PostgreSQL · ChromaDB · Docker · GitHub Actions — github.com/venkatalakshmiparimala/AgentProbe

- 15-endpoint **FastAPI backend** with **Celery async task workers**, ChromaDB vector store, and REST API design; **data ingestion pipelines** for embeddings and retrieval; Docker containerized; GitHub Actions CI/CD; ships to real users. Designed for **independent end-to-end ownership** - one engineer, full stack, production-grade.

Leasel — Backend AI Platform with Data Ingestion Pipeline

Backend · Data Pipelines · PostgreSQL

Python · FastAPI · LangGraph · PostgreSQL · SQLite · OCR · Docker — github.com/venkatalakshmiparimala/Lease_complaince_rag

- Production backend with **end-to-end data ingestion pipeline**: OCR document parsing, section-aware chunking, hybrid search indexing, and **PostgreSQL/SQLite storage** over 1,240 documents; FastAPI REST API layer; async task orchestration; CI eval harness: **93.6% precision, 87.9% F1**. Entirely owner-operated, no handoffs.

AI on FHIR — Full-Stack with Django-Style Auth & REST APIs

Full-Stack · REST APIs · RBAC

Python · Flask · React · TypeScript · PostgreSQL · Docker · REST APIs · OAuth2 · JWT — github.com/venkatalakshmiparimala/AI_on_FHIR

- Full-stack (React/TypeScript frontend + Python backend): **RESTful API design, OAuth2/JWT security, HIPAA-style RBAC**; PostgreSQL data modeling; Docker containerized; cut manual lookup time ~60%. Demonstrates **comfortable owning the full stack independently**.

EDUCATION

M.S. Data Analytics Engineering — George Mason University — Graduated May 2026

GPA: 3.87 · Fairfax, Virginia · Distributed systems, ML pipelines, data governance, database design, software architecture

B.Tech Computer Science & Engineering — Jawaharlal Nehru Technological University, Anantapur — 2013 – 2017